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Rotation Matrix	Translation Matrix	Offset Matrix	Test Vector
$R := \begin{pmatrix} r_{11} & r_{12} & 0 \\ r_{21} & r_{22} & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$T := \begin{pmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{pmatrix}$	$T_{\text{ofs}} := \begin{pmatrix} 1 & 0 & -t_x \\ 0 & 1 & -t_y \\ 0 & 0 & 1 \end{pmatrix}$	$V := \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$

Rotate a vector 2x2

$$R \cdot V \rightarrow \begin{pmatrix} r_{11} \cdot x + r_{12} \cdot y \\ r_{21} \cdot x + r_{22} \cdot y \\ 1 \end{pmatrix}$$

Translate a vector 2x2

$$T \cdot V \rightarrow \begin{pmatrix} t_x + x \\ t_y + y \\ 1 \end{pmatrix}$$

Rotate then translate

$$T \cdot R \rightarrow \begin{pmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{pmatrix}$$

Translate then rotate

$$R \cdot T \rightarrow \begin{pmatrix} r_{11} & r_{12} & r_{11} \cdot t_x + r_{12} \cdot t_y \\ r_{21} & r_{22} & r_{21} \cdot t_x + r_{22} \cdot t_y \\ 0 & 0 & 1 \end{pmatrix}$$

Translate then rotate then translate (rotate around the origin (t_x, t_y))

$$T \cdot R \cdot T_{\text{ofs}} \rightarrow \begin{pmatrix} r_{11} & r_{12} & t_x - r_{11} \cdot t_x - r_{12} \cdot t_y \\ r_{21} & r_{22} & t_y - r_{21} \cdot t_x - r_{22} \cdot t_y \\ 0 & 0 & 1 \end{pmatrix}$$